Project Proposal

ADP Program	ADP CS (Computer Science)
Project ID *(for office use)*	
Title of Project	**Admission Management System (AMS)**
Type of project	[ ] Traditional [✓] Industrial [ ] Continuing
Nature of project	[✓] **D**evelopment [ ] **R**esearch [ ] **S**urvey
Area of specialization/ Field	Education / Academic Administration

Project Group Members

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Declaration: Project group members have cleared all prerequisites courses for project as per their degree requirements.

Supervisor Name and Signatures: **Principal:**

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1. Introduction

In the modern educational landscape, institutions face significant challenges in managing admissions, particularly when dealing with high volumes of applicants. The Admission Management System (AMS) is developed to address these challenges by replacing outdated, paper-based methods with a streamlined, digital solution. This project focuses on reducing administrative workload, preventing errors, and improving transparency by enabling students to complete applications and submit necessary documents online. AMS offers a user-friendly interface that allows students to track their application progress, while providing administrators with efficient tools to review and process applications. By implementing AMS, educational institutions can modernize their admission workflows, enhance communication with applicants, and deliver a faster, more organized admission experience.

1.2 **Background of the Project**

In the current academic environment, institutions often struggle with the complexity of managing admissions, from handling numerous applications to maintaining clear and transparent processes. The Admission Management System (AMS) seeks to solve these challenges by providing a digital platform that simplifies application submission, reduces manual paperwork, and improves communication between students and administrators.

1.3 **Problem Statement**

Manual admission processes create challenges for both students and administrators. Paper-based systems can cause data entry mistakes, processing delays, and confusion about application status, leading to frustration and inefficiency throughout the admission cycle.

1.4 **Purpose of the System**

AMS offers an integrated web-based solution where students can easily apply for admission, attach required documents, monitor the progress of their applications, and stay in touch with administrators during every stage of the process.

1.5 **Target Users**

Students, Admission Officers, and Administrators

2. Objective

The goal of this project is to create an online admission system that makes it easier for students to apply for admission and for administrators to handle and review those applications. It will be a secure, role-based system where students can fill out forms and upload documents, and admins can check and process applications quickly and easily.

2.2 **Main Objective**

To develop a web-based admission management system that enables students to register, submit applications, upload required documents, and track their application status in real-time, while providing administrators with tools to review, manage, and process applications efficiently within a secure digital environment.

2.3 **Sub-Objectives**

- Enabling secure login and signup for students and administrators
- Allowing students to submit applications and upload required documents
- Providing admin controls for managing, reviewing, and processing applications
- Offering an intuitive user interface for a seamless admission process experience

3. Project Scope

The scope of this project is limited to the development of a web-based system that offers basic functionalities such as student registration, application form submission, document upload, role-based login, and application status tracking. The system will include two types of users: administrators and students. Each user type will have access to a personalized dashboard and relevant features.
The system assumes users have reliable internet connectivity and use up-to-date web browsers. It does not include advanced functionalities such as scholarship management, automated interview scheduling, fee payment integration, or connections to external academic databases. At this stage, the project is focused on creating a straightforward, secure, and efficient admission process that simplifies application handling for both students and administrators.

3.2 **In-Scope Features**

- Student login/signup and profile management
- Online application submission and document uploads
- Application status tracking and notifications
- Chatbot
- Challan Generate
- Scholarship flow
- Role-based authentication and access control, User portal and Admin Portal

3.3 **Out-of-Scope Features**

- Online fee payment processing (future enhancement)
- Integration with external academic or examination systems (future enhancement)

3.4 **Assumptions**

- All users have stable internet access
- Students provide accurate and truthful information in their applications
- The platform is deployed on reliable cloud services

3.3 **Limitations**

- No integration with external educational or examination board systems

3.4 **Programming Languages**

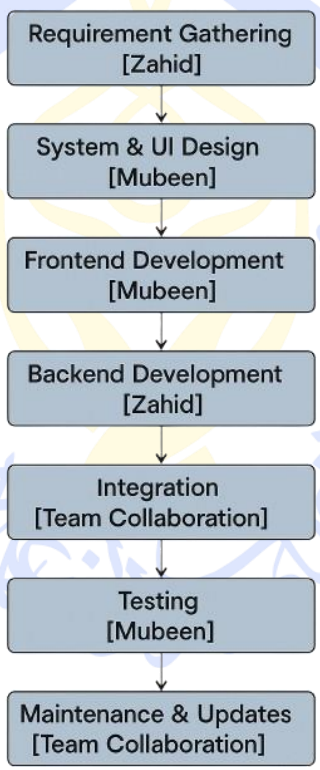
The AMS is built using the following technologies:

- **Frontend:** React (JavaScript/JSX), Tailwind CSS, HTML5, CSS3
- **Backend:** NestJS (TypeScript), JWT for authentication
- **Database:** MongoDB with Mongoose ORM

4. Summary

AMS is a full-stack web application that provides a digital solution for managing student admissions. It features two primary user roles: admin and student. Administrators manage applications and review submitted documents, while students register, submit applications, and track their admission status. Built with React (frontend), NestJS (backend), and MongoDB with Mongoose (database layer), the system uses JWT (JSON Web Tokens) for secure authentication and role-based access. The project aims to streamline the admission process, reduce manual paperwork, and improve communication between applicants and administrators.

4.2 Word Breakdown Structure:



4.2 Phases and Milestones

- TASK 1: Requirement Gathering and Analysis

- TASK 2: UI Design and Environment Setup
- TASK 3: Logical System Design
- TASK 4: Frontend Development and Integration
- TASK 5: Backend Development (Student and User Modules)
- TASK 6: Backend Development (Admin Panel and Management Modules)
- TASK 7: Database Design and Integration using MongoDB and Mongoose
- TASK 8: Testing, Bug Fixing, and Deployment

4.3 Gantt Chart

Phase	Week 1	Week 2	Week 3	Week4	Week 5	Week 6	Week 7	Week 8	Week9
Planning & Design									
Development									
Testing & Launce									